



WESTMINSTER ASTRONOMICAL SOCIETY, INC. (WASI)

★
Membership News



Volume 42 – Number 2

Spring 2026

Message from our Society's President, Brooke Parkhurst

My first visit to a WASI event was a revelation. Everyone was so welcoming, and so happy to just geek out about stars and share their (considerable!) knowledge. What a lovely place to be!

In a time when it is more important than ever to reaffirm the validity of scientific expertise and the real importance of science in our everyday lives, I'm so proud to be part of this group that does those things with such grace, kindness, and enjoyment!

Keep it up!

Notes and Thoughts

This newsletter goes out quarterly. Do you have thoughts, insight or useful information for the members? Please consider submitting an article. Our survey last year showed that members really want more learning resources, so your expertise could be valuable to many. Send your ideas to secretary@westminsterastro.org.

Did you have to take a required minimum distribution on investment accounts, or did your tax return have a nice surprise in terms of a refund? WASI is a 501 (c)(3) organization, so any gift is tax deductible. Contact treasurer@westminsterastro.org for more information.

Did you know that Night Sky Network publishes a monthly article that connects what we see in the sky to NASA science? They are released by the 20th of the previous month. Find out more here: <https://bit.ly/nightskynotes>.

Fun fact: Saturn's rings are 150,000 km in diameter, yet only 10 meters to 1 km thick.

More fun: The Oort cloud, which is a hollow sphere surrounding our Solar System, contains about 1 *trillion* comets.

And get this: Evan Dincher's picture of M42, the Great Orion Nebula, is featured later in this newsletter. It looks dense but there's only about 1000 atoms of gas per cubic centimeter. Compare that to Earth's atmosphere at sea level, where there's 50,000,000,000,000,000 atoms per cm³.

To Pixinsight or Not to Pixinsight

A lot of folks pursuing astrophotography are reluctant to start using the PixInsight software program (<https://pixinsight.com/>) as it has the reputation of being nearly inscrutable. Is it really that hard to use?

Astrophotographers take many (often hundreds or even thousands) of short-ish exposures of the celestial body we're imaging. Later, snug at home in front of a computer, we use software like Astro Pixel Processor, Siril, AstroArt or PixInsight to combine those short snaps into a single image and then process it to bring out the picture's beauty. That involves reducing noise, removing light pollution and other image defects, cleaning up fuzzy stars, and much more. Some of these programs, like AstroArt, are very simple. Others, less so. Many astrophotographers eventually migrate to PixInsight as it offers more capability than any other program.

But it sure has a bad rep! People complain you need to know too much math, or that the user interface is hopelessly arcane. What's the truth?

Arcane it is. There's frustratingly-little consistency between the tools. Practice is needed. One needs to cultivate a stable of the PI tools you use most frequently and keep those on the PI desktop.

If you are not interested in spending hours in front of a computer making a silk purse from, well, let's just say not-the-best raw images, fuhgetaboutit. You'll hate most processing software and will consign PixInsight to a special place in Hades. It offers so many options, and is so very slow in some processing modes, that, for some, the frustration level will be high. It is said that if there are three ways to do something, PixInsight gives you 15 tools to do that.

If your computer isn't reasonably fast the program can be nearly unusable. Some people mix programs, using, for instance, Siril for stacking and Pixinsight for everything else. The key is a fast computer, with lots of RAM (64 GB or more), a microprocessor with as many cores as you can afford, and gobs of very fast semiconductor disk (NVMe).

Then there's the money. PixInsight is 300 Euros. Two add-ons (BlurXterminator and NoiseXterminator) at another \$160 are so useful they are nearly mandatory. Many use Adam Block's video tutorials to learn the program; that's another non-trivial investment.

PixInsight is almost an operating system for image processing. The main screen offers some basic functionality to manage files, stretch images, zoom and the like, but its power comes from what are called processes and scripts. The two are very similar and comprise hundreds of individual tools. Some reduce noise. Some combine narrowband images into a single result. Others evaluate the image quality.

It is impossible to learn PixInsight from the resources that come with it. There is no manual. Some, maybe half, of the tools have documentation. Unfortunately, that is uneven and can be hard to understand, often overflowing with difficult math. If you like math the docs are fascinating, but in general you simply don't need to understand all of the nuances. Instead, use the resources listed later in this article.

To Pixinsight or to Not Pixinsight—Continued

Part of what puts people off from PI is that each tool often has dozens of parameters that must be set. For example, this is just a portion of the setup for the tool that stacks individual frames together into a single image:

The screenshot displays the Pixinsight software interface, specifically the 'Image Integration' and 'Pixel Rejection' tool settings. The 'Image Integration' panel includes dropdown menus for 'Combination' (set to 'Average'), 'Normalization' (set to 'Additive with scaling'), and 'Weights' (set to 'PSF Signal Weight'). It also features input fields for 'Weight keyword', 'CSV weights file', and 'Minimum weight' (0.005000). A series of checkboxes allows users to select processing options: 'Generate integrated image' (checked), 'Generate a 64-bit result image' (unchecked), 'Generate drizzle data' (unchecked), 'Subtract pedestals' (unchecked), 'Truncate on out-of-range' (unchecked), 'Close previous images' (unchecked), and 'Automatic buffer sizes' (checked). Below these are sliders for 'Buffer size (MiB)' (16) and 'Stack size (MiB)' (1024), along with a checked 'Use file cache' option. The 'Pixel Rejection (1)' panel shows the 'Rejection algorithm' set to 'Generalized Extreme Studentized Deviate (ESD)' and 'Normalization' set to 'Scale + zero offset'. It includes checkboxes for 'Generate rejection maps' (checked), 'Clip low pixels' (checked), 'Clip high pixels' (checked), 'Clip low range' (checked), 'Clip high range' (unchecked), 'Report range rejection' (unchecked), and 'Map range rejection' (checked). The 'Pixel Rejection (2)' panel contains numerous sliders for various parameters: 'Min/Max low' (1), 'Min/Max high' (1), 'Percentile low' (0.200), 'Percentile high' (0.100), 'Sigma low' (4.000), 'Sigma high' (3.000), 'Winsorization cutoff' (5.000), 'Linear fit low' (5.000), 'Linear fit high' (4.000), 'ESD outliers' (0.30), 'ESD significance' (0.05), 'RCR limit' (0.10), 'Range low' (0.000000), and 'Range high' (0.980000). The interface uses a light gray theme with blue headers for each section.

Format Hints

Image Integration

Combination: Average

Normalization: Additive with scaling

Weights: PSF Signal Weight

Weight keyword:

CSV weights file:

Minimum weight: 0.005000

☒ Generate integrated image

☐ Generate a 64-bit result image

☐ Generate drizzle data

☐ Subtract pedestals

☐ Truncate on out-of-range

☐ Close previous images

☒ Automatic buffer sizes

Buffer size (MiB): 16

Stack size (MiB): 1024

☒ Use file cache

Pixel Rejection (1)

Rejection algorithm: Generalized Extreme Studentized Deviate (ESD)

Normalization: Scale + zero offset

☒ Generate rejection maps

☒ Clip low pixels

☒ Clip high pixels

☒ Clip low range

☐ Clip high range

☐ Report range rejection

☒ Map range rejection

Pixel Rejection (2)

Min/Max low: 1

Min/Max high: 1

Percentile low: 0.200

Percentile high: 0.100

Sigma low: 4.000

Sigma high: 3.000

Winsorization cutoff: 5.000

Linear fit low: 5.000

Linear fit high: 4.000

ESD outliers: 0.30

ESD significance: 0.05

RCR limit: 0.10

Range low: 0.000000

Range high: 0.980000

Large-Scale Pixel Rejection

Scary! But the references listed on the next page tell you what settings to use. And the default settings are generally just fine.

To Pixinsight or to Not Pixinsight—Continued

I said that if you don't like hours wrestling with the computer PI is not for you. As you get more comfortable with the program, and perhaps more curious, you may find it worthwhile to explore the options to learn how to tweak them to optimize your images. I find this a voyage of discovery taking me all over the web to learn what these parameters mean. It's pretty cool.

As with all astro image processing, there are two basic steps. In *preprocessing* we calibrate the images, remove things like Elon Musk's satellites (Muskellites?), do noise reduction, align all of the images, and then stack them into a single image, or one per filter. *Postprocessing* is taking those one or few images, stretching them to pull data from the darkness, and cleaning them up, assigning colors (if you're imaging with narrow-band filters), adjusting colors, and generally putting a final pretty spin on the result.

Preprocessing can be quite tedious, but PI offers a tool (Weighted Batch PreProcessing) that makes this brain-dead simple. Your computer may grind for hours, but it's not unusual to start the tool before going to bed. The other day I preprocessed 1925 subs on a very fast computer: It took 15 hours.

Postprocessing is where you put your stamp on the image and can be quite creative. You'll use a lot of PI's tools, and here's where you'll really want to learn the program. Figure on hours or days of work, depending on the image and how fussy you care to be. But the bottom line is that PI can help produce some wonderful images.

Pixinsight is frustrating and demanding. But using it is a ton of fun and the results: spectacular.

If you're interested in a WASI event to introduce the basics of Pixinsight, email secretary@westminsterastro.org. If there are at least a few responses we'll set up a class.

Pixinsight Learning Resources

Pixinsight changes constantly. Other than Adam Block's videos all of these resources are out of date, but are correct enough to get you going.

The Deep-Sky Imaging Primer (\$50) by Charles Bracken. This is possibly the best book for a general intro to astrophotography. It's easy to understand but does not delve deeply into Pixinsight. There's enough in the book to get you started and making a decent image. But only started; you'll want to use the other resources listed here.

Mastering PixInsight (\$27) by Rogelio Andreo is a pair of PDF files you need and will want to print (though comprise about 700 pages). It's a little uneven in that some descriptions are word-for-word from PI's tooltips, but this is the best single reference. Buy it here: <https://deepskycolors.com/books/mastering-pixinsight/>.

My Pixinsight Workflow (free PDF) by Christopher Foster (https://drive.google.com/drive/folders/1jXtkjWotcn0Is-m_DZyTgim5wxFCUOgU) is an excellent and practical how-to guide.

Adam Block (<https://www.adamblockstudios.com/>) offers online video courses. They are very good but the pace is mind-numbingly slow—run them at 2x speed. He offers so many classes and options that it can be confusing. You can spend quite a bit of money and time on these.

Observing Quasars

What is the furthest object in the universe you can see?

Generally, with naked eyes the Andromeda galaxy is about the most distant object one can see, at over 2 million light years away. Some can pick out M33, the Triangulum Galaxy, which is a bit further. Those distances are unimaginably vast... but these objects are in our Local Group, and are nearby neighbors.

A big telescope pushes that horizon to much larger distances. The galaxies in the Coma Cluster, 300 million light years away, can be seen with a decent scope.

Astrophotography changes things. Very, very faint and distant objects can be detected with modest equipment.

Some of the most distant objects are quasars (aka QSO). These are supermassive black holes eating the heart out of galaxies. The energy released by material being torn apart by the black holes is staggering. They're bright, cosmically speaking, and some are not hard to capture with long-exposures.

Bright, but small. Their light varies over hours to weeks which means they can't be more than a few light-weeks in diameter.

The most distant QSO detected by the pros is almost 32 billion light years away from us. We amateurs will never see it. With a red shift of 10.1 its photons are in the infrared to the radio spectrum. This object's light has been traveling since less than 500 million years after the formation of the universe.

Consider 3C273, a QSO about 2.4 *billion* light years distant. Its apparent magnitude is 12.8, well within reach of pretty much any astrophotography rig. It looks like a dim star... but how cool to pick up such a distant object!

And how bright is it? Its absolute magnitude is -26.7 . If it were 33 light years away from us, eight light years further than Vega, it would be as bright as the midday sun. Think about that!

If you're doing long exposures you can go much further. Recently, while processing an image of Abell 779, I hunted for very dim QSOs, finding SDSS J091830.41+334824.4, a name that fairly rolls off the tongue. (Perhaps Frank Zappa named one of his kids after this). That puppy is at a red shift of 3.9, which puts it (today) 24 billion light years away. The light my camera captured left the QSO 12.3 billion years ago, photons in lonely traverse since just 1.5 billion years after the Big Bang. At magnitude 21.7 it's millions of times fainter than any naked eye object, despite being as bright as several thousand galaxies. In the image it's barely noticeable, a myopically-challenging few pixels of light.

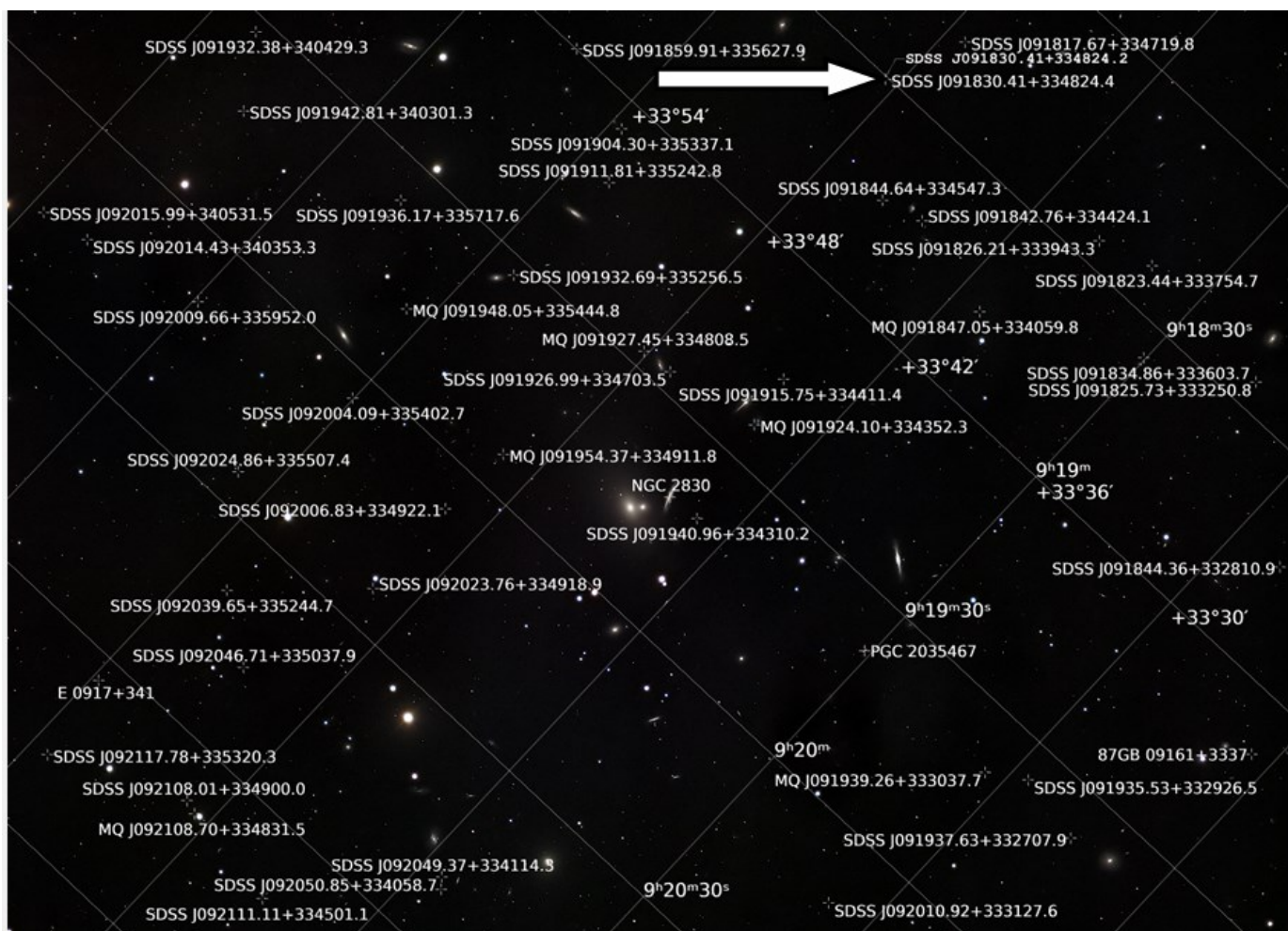
Various tools will annotate an astrophoto with the names of the objects in the image. If you enable the million-quasar database many of these pictures will be covered with QSOs. The annotations show where there are QSOs, though your image may show nothing if the object is too dim.

Observing Quasars, Continued

But... why can we see these objects at all? A great deal of their energy is from what's called a Lyman Alpha electron transition. Do you image with narrowband filters? If so, you're familiar with the hydrogen-alpha filter, which only passes light at 656 nanometers. Lyman-alpha transitions are similar, but radiate at 122 nm, which is far into the ultraviolet, undetectable by our eyes and cameras.

Since the universe is expanding, the more distant an object is, the faster it's moving away from us. This streeeeetttches out the wavelength of photons, an effect called red shift, since such stretching shifts the light from the UV to the visible part of the spectrum. With a redshift of 3.9, the arrowed QSO in the image below emits light at 122 nm, but those photons are shifted to visible light.

The first quasars were detected in the 1950s but weren't really understood till the 70s. How cool it is that we can pick up these early artifacts of our universe with our amateur equipment!



Abell 779, a galaxy cluster in Lynx. Every annotation with a long name depicts a quasar, though some are so dim the annotation is there, but the QSO isn't visible. There are a lot of quasars in this sub-one-degree field of view! The arrow points to one that is currently 24 billion light years away from us.

The Bortle Scale

Oh, how we love to look up at the night sky and see... a bright streetlight. The reflected glare of Reisterstown. Headlights bouncing off RT 140.

There are plenty of ways to rate the visibility of stars, ranging from the qualitative WOW to all-I-see-is-skyglow. Or, you can buy meters which will give a quantitative assessment using the SQM scale. That measure is brightness in magnitudes per arcsecond squared. Squared, because this is measured over an area. An SQM of 21.7 to 22.0 means astronomy heaven. 21ish is rural, 20.8 to 21.3 you're losing stars, and for those unfortunates in skies under 19 or so it probably pays to take up stamp collecting. (If you'd like to get a sky quality meter, check out the discussion of them here: <https://www.cloudynights.com/forums/topic/923523-sky-quality-meter-recommendations/>.)

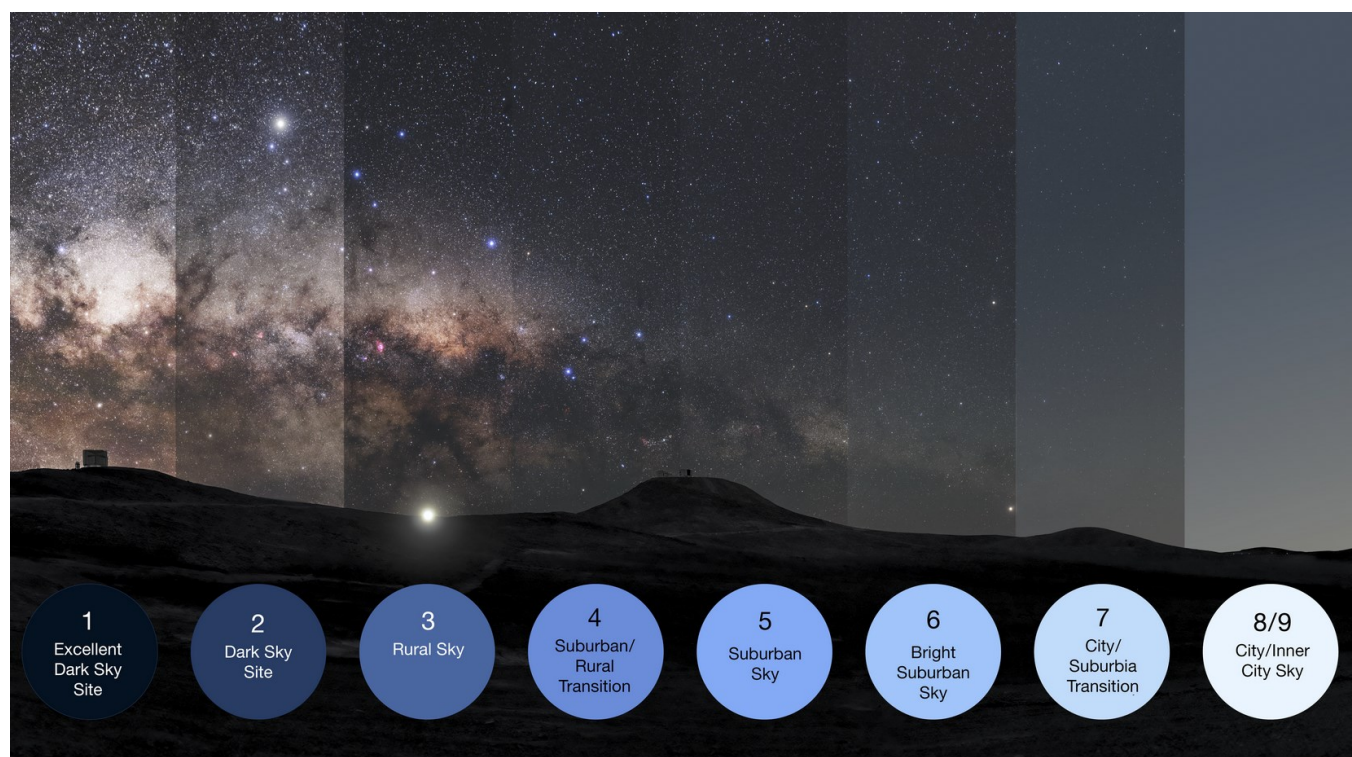
Alternatively, there's naked eye limiting magnitude (NELM) which indicates the magnitude of the dimmest star one can see.

25 years ago John Bortle came up with a simpler scale which ranges from 1 (perfect skies) to 9 (Baltimore). Bortle 1 is a NELM of an incredible nearly 8, where Bortle 9 corresponds to worse than magnitude 4—maybe you can pick out the Pleiades.

You can view an interactive light pollution map here: <https://www.lightpollutionmap.info/>. Click on any point and it will display the sky's statistics. Alas, here in Maryland it's hard to find anywhere much better than Bortle 4 (that's the number the map gives for Bear Branch Nature Center).

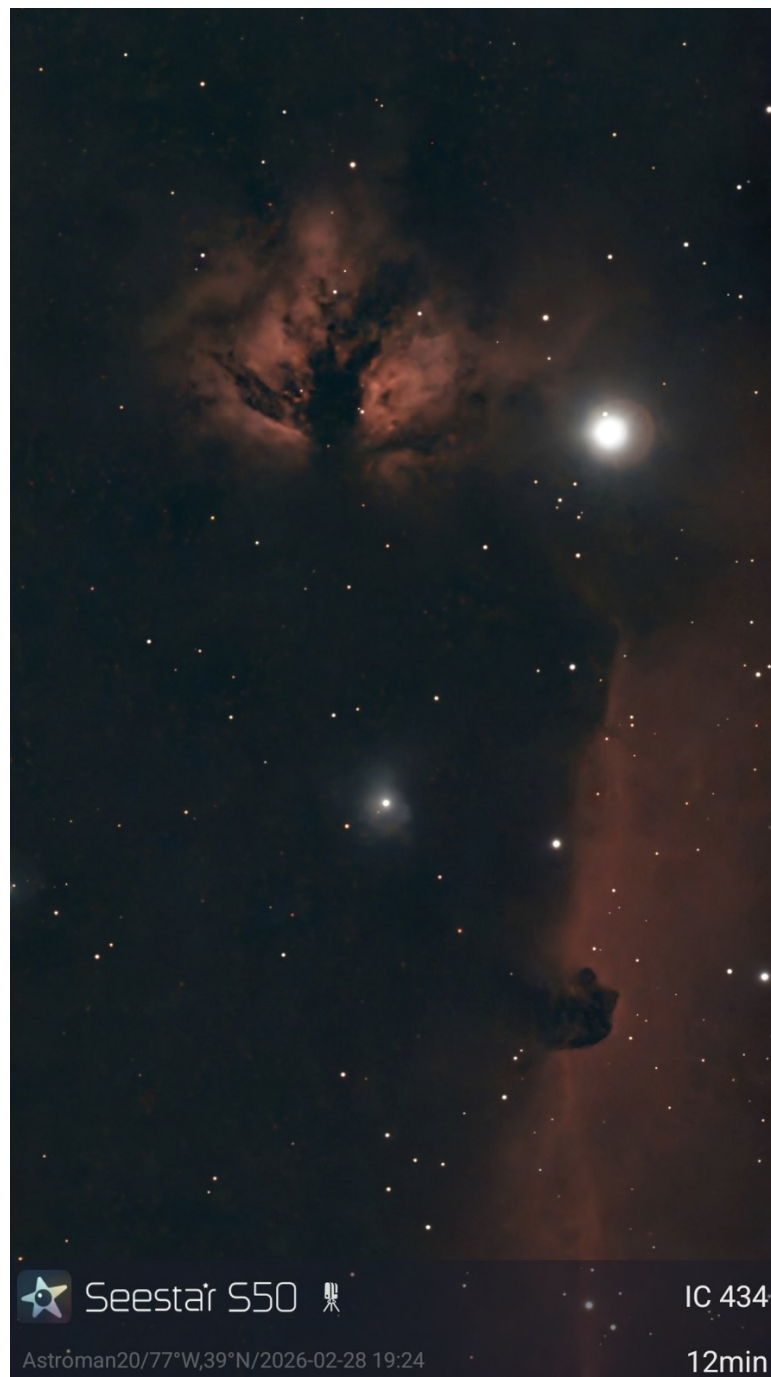
Dark Sky International (<https://darksky.org/>) fights light pollution and has good resources. The sad truth is, though, that it's estimated skies get about 10% worse every year.

Some wags suggest that the best bit of astronomy gear is a tank of gas to get away from the city.



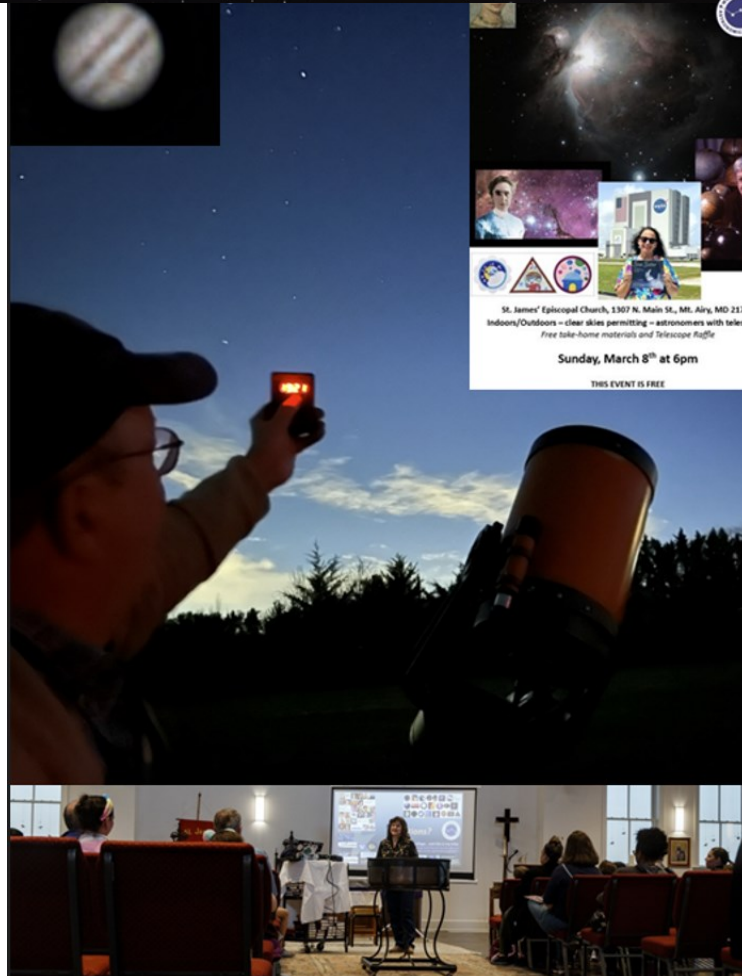
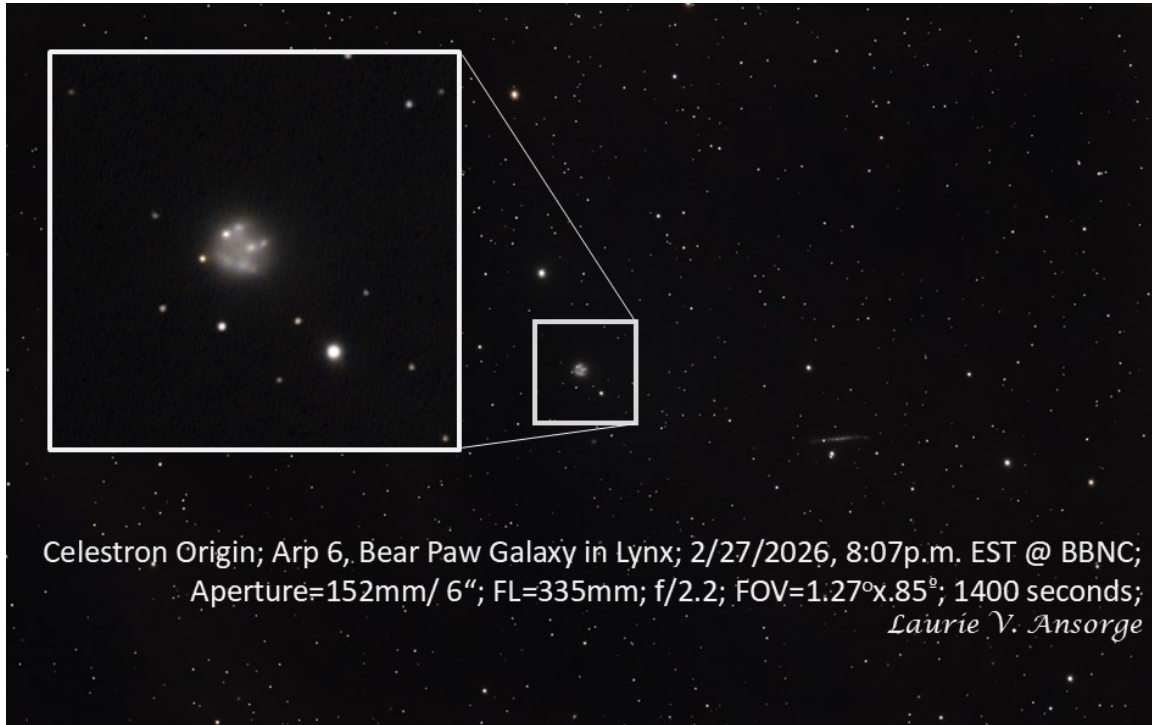
Astrophotos From Our Members

Evan Dincher has been getting great pictures with his Seestar 50. M42 is a perennial favorite, and the nearby IC434 shows the Horsehead Nebula. Note the exposure times—that S50 gets some great results with surprisingly short exposures.



Astrophotos From Our Members

Laurie Ansorge captured the Bear Claw Galaxy and other objects:



Astrophotos From Our Members

Matt Orsie wrote: "On 2-8/9-26 between 7:15pm and 2:00am I grabbed a mosaic of NGC 2237 with my Seestar S50. Being a large nebula, larger than the Seestar S50's sensor, I used mosaic mode at 1.7x. Mosaic mode will have many more dropped subs because of it's constant panning to encompass the selected region to photograph (I had about 11% dropped subs) getting this larger target area. In the process of capturing the large area the central region gets more capture time than the periphery so there is more noise on the edges. That's the trade off with the Seestar S50 in mosaic mode. Siril and Photoshop."



Astrophotos From Our Members

Matt also sent this: "Here's a 7hr 15min stack of M81 and M82 from March 8. I used the Seestar's dual band filter to get 3 hrs of HII and the IR/Cut filter (default) to get 5 hrs of broadband. This is a 3x drizzle. Siril and Photoshop."



Astrophotos From Our Members

Konstantin Vishnevsky is another happy Seestar user: “I just re-processed the NGC 2359 (Thor's Helmet). Captured with Seestar S50 and processed in PI with the new MultiscaleAdaptiveStretch, BXT, NXT and SXT.”



Astrophotos From Our Members

Alan Goldberg wrote: “Two images taken by my S50 last night (March 9-10) from Brookeville. I don’t claim to take them, because the S50’s automation does all the work. I just crop and adjust the result on my iPhone. I have yet to process any individual subs.

M1 is a 64 min. Integration.

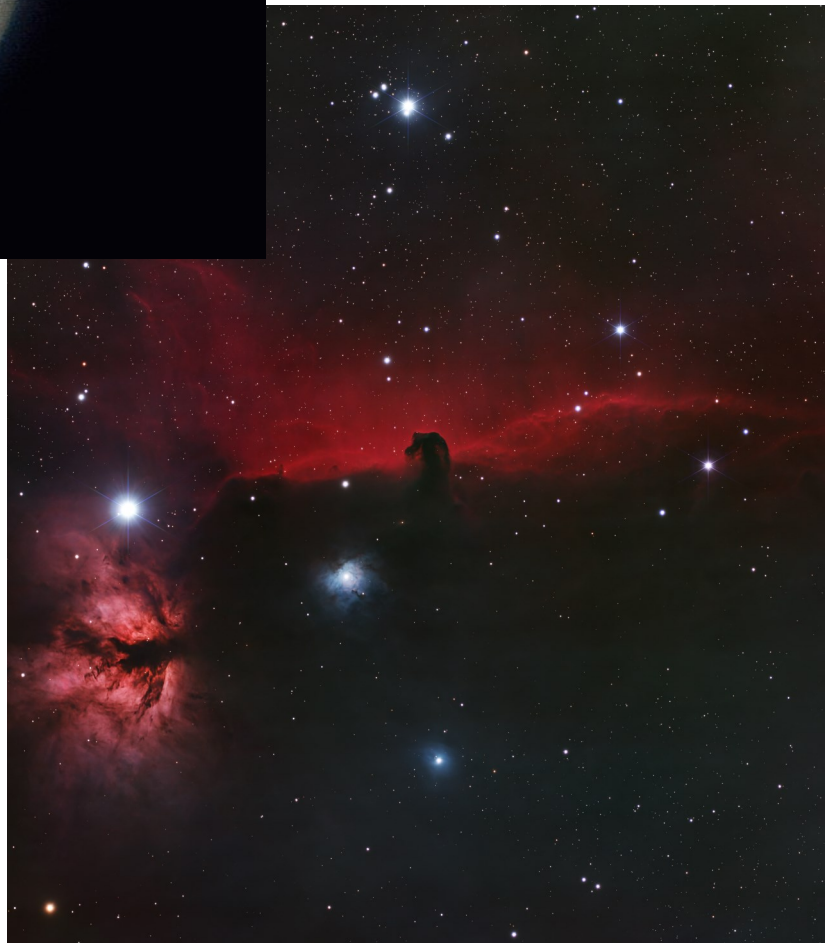
The Leo Triplet is 101 min total integration, automatically mosaicked at 120% of a full frame.”



Astrophotos From Our Members

Dave Weisman sent these. Jupiter was taken March 1 and you can see IO and its shadow. Taken with a Celestron Edge 9.25 and ZWO planetary camera, processed in Wave Sharp. Dave sent this as an animated .gif; alas, we can't show the animation here but it's amazing to watch IO moving.

The Horsehead and Flame Nebulas were taken with an Askar 500 with about 3.5 hours of integration with a ZWO 2600MM, processed in HRGB with Pixinsight.



WASI FAQs

Library - Did you know we have over 700 books about astronomy in our WASI library? There are available to WASI members. Here's the complete card catalog: <https://westminsterastro.groups.io/g/main/files>.

Loaner telescopes - We also have a telescope lending library. If you'd like to borrow a scope, talk to Curt Roelle.

Astronomical League - All WASI members are also members of the Astronomical League. Check out their 80+ observing programs, many of which come with awards: <https://www.astroleague.org/>

Newsletter - Send pictures, articles, and ideas for the newsletter to secretary@westminsterastro.org.

Facebook - We're active and sharing images on our Facebook page, found here:



Join/Renew membership link: <https://www.westminsterastro.org/join-wasi/>

If you've already entered your contact information (renewing), skip the "database" link: <https://paypal.me/WAstroSInc>

Dues are payable via PayPal on the link above, by check or cash (and through your bank's on-line bill payment). Membership Dues are \$25/year for individuals or family, and youth under 18 is \$5/year. Or, save money and sign up for 2 years (\$40), or 4 years (\$80).

- On time payment means eligibility for the annual incentive.
- Access to the club's observatory (after training).
- Keep access to the members-only groups.io pages/information
- Receive members-only access/notifications on Night Sky Network
- Keep/get discount rates for popular astronomy magazines
- Borrow from the WASI scope/literature library

Files and club member correspondence & wiki links are found here: <https://westminsterastro.groups.io/g/main>. Remember to set your communication preferences.

Outreach/event calendar is found on: <https://nightsky.jpl.nasa.gov/index.cfm>. Set your communication preferences here as well.

Changed address, email or phone? Please update your information and send a message to the webmaster and/or treasurer@westminsterastro.org.

We meet monthly on the 2nd Wednesday of the month:
Back to Basics from 7:00 PM – 7:30PM; General Meeting 7:30PM – 9:30PM
Bear Branch Nature Center Carroll County; 300 John Owings Rd.; Westminster, MD 21158
Website: <https://www.westminsterastro.org/> (Zoom info for hybrid meetings)